

◆ 1. Momentum Method Numerical

✚ Given:

- Learning rate $\eta = 0.5$
 - Momentum $\alpha = 0.8$
 - Error term $\delta = 0.6$
 - Input $x = 1$
 - Previous weight change $\Delta w(t - 1) = 0.2$
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◆ Formula:

$$\Delta w(t) = \eta \delta x + \alpha \Delta w(t - 1)$$

◆ Step-by-Step Solution

Step 1: Compute Gradient Term

$$\eta \delta x = 0.5 \times 0.6 \times 1 = 0.3$$

Step 2: Compute Momentum Term

$$\alpha \Delta w(t - 1) = 0.8 \times 0.2 = 0.16$$

Step 3: Add Both Terms

$$\Delta w(t) = 0.3 + 0.16 = 0.46$$

✅ Final Answer:

$$\Delta w(t) = 0.46$$

◆ Weight Update (Extra Step)

If current weight $w = 1.0$:

$$w_{new} = 1.0 + 0.46 = 1.46$$

◆ 2. Momentum Method (Next Iteration)

📌 **Given:**

Now previous update becomes:

$$\Delta w(t - 1) = 0.46$$

Step:

$$\begin{aligned}\Delta w(t) &= (0.5 \times 0.6 \times 1) + (0.8 \times 0.46) \\ &= 0.3 + 0.368 = 0.668\end{aligned}$$

✅ **Final:**

$$\Delta w = 0.668$$

👉 Shows **acceleration effect** (update increases)